# Machine Learning Spring 2025/2026
## *Project Guidelines*

This document presents some guidelines about the project you are going to do for the Machine Learning course. It basically focuses on four main points

*(1) the source for datasets*
*(2) the project format*
*(3) the expected outcome*
*(4) and the deadlines.*

**Projects must be done in groups of 2 students** (single person project are allowed only motivating the request).

## Datasets

To find a dataset for your project, you can exploit the UCI Machine Learning Repository: https://archive.ics.uci.edu/datasets or Kaggle https://www.kaggle.com/datasets or any source you may identify as adapt for the specific activity you are targeting. In this repository you will find hundreds of datasets about a large variety of domains. Go through a subset of these datasets and choose the one that inspires you most or the one that is in line with your work or research activities. For each dataset you can find a reference paper that can help you to better understand it.

If you have an idea about what you want to do, you can rely on two main factors that can help you to choose a dataset for your project.
The first factor could be the type of tasks you want to do (e.g., regression, classification, clustering, or reinforcement – see also the beta version of the website). Based on that you can filter out unsuitable datasets for your task.
The second factor could be the dataset itself. You might have a preference for numeric features, textual features, or data with network structures. By contrast, if you do not have an idea about what you want to do, then you just choose based on the domain that you like most, then you have to discover by yourself what could be done with the chosen dataset. For that, look carefully at the characteristic of the dataset to be able to choose the right task for it.

## Project Format

Your project should contain a complete solution to a given problem. A complete solution covers 4 main parts:

*(1) Problem Description: the format of the training data and the list of tasks to be achieved specifying the training algorithms to be used*
*(2) Feature Selection: how features are chosen? in case a subset is chosen to solve your task describe the model selection strategy*
*(3) Algorithm Implementation*
*(4) Results: the experiments that validate your work.*

Depending on your interests and background, you can choose one of the following formats for your project. You can also decide to combine more formats.

***Format 1.***

In this format, the **focus of the project would be on data processing and preparation**.

Possible directions could be (1) studying feature selection strategies through extensive experiments. You can choose to use well established measures like information gain (IG), mutual information (MI), etc. Alternatively, you can choose to discover new features by extracting more information from data, for example using frequent pattern mining (2) distant learning which aims at automatically labelling data and thus creating a training dataset with less human effort. The point here is for you to learn the strategy to follow to be able to achieve this task. Note that your project must cover all the 4 parts described above. Since most of the work, if you choose this format, is done at the data processing part, you can just use existing packages to run some learning algorithms and observe the results. **The experiments you do and the results you get should be about the impact of your data processing techniques (feature selection or distant learning) on the learning results.**

***Format 2.***

In this format, the **focus of the project would be on the learning algorithms.** The goal is to run several algorithms to achieve the same task using the same dataset, and compare their performances. This format is suitable for students how do not have a strong background in the filed. So, the goal is to learn how the chosen machine learning algorithms work and how they behave. To have a meaningful study, you should use learning algorithms that are fundamentally different. For example, one mathematical model like SVM, one statistical model like Naive Bayes, one rule-based model like random forests, and one deep learning model like neural networks. Your task would then be to run these algorithms and observe the results, by tuning their parameters. The results should contain your observations and interpretations. **The expected outcome is that you can give some explanations why a certain algorithm is behaving good or bad for the specified task. That could be related to the dataset structure, the size of the dataset, or the limitations of model itself. In this format, you do not need to implement the algorithms but use existing tools.**

***Format 3.***

In this format, the **focus of the project would be on deeply understanding a given learning model.**

You can choose an algorithm and implement it yourself using the programming language you feel most comfortable with. This format is used if you really want to become an expert on a specific model. Only by implementing the whole approach and putting all the steps together you can understand what an algorithm is really doing. It is important to note that it is not expected for your algorithm to run as fast as existing algorithms that you find in machine learning tools. The existing tools use advanced optimization techniques which are the result of years of work. **The project of this format would contain all the 4**

**parts with an extensive section about the implementation, the encountered problems and the solutions found. The experiments should contain a comparison of your algorithm and an existing implementation (baseline). Both your algorithm and the baseline should run on the same dataset with same (comparable) parameters and the results should show the effectiveness of both.** Then, you should describe what lessons you have learned from the work you have achieved.

**Format 4.**
In this format, the **focus would be on the results.**
In other words how to improve the results of a machine learning algorithm. This can be done at two levels. First, at the level of effectiveness, means how can you make the results more accurate. To address this problem, you can use some standard strategies, like bagging or boosting which involve the usage of several learning models. Alternatively, you can decide to combine several learning paradigms like supervised and unsupervised learning techniques. For example we can cluster the data and then run the supervised learning on each cluster to see the impact on the results. Second, at the level of efficiency. This can also be done by combining supervised and unsupervised techniques. For example if an algorithm runs slow on the whole dataset we can use clustering to find representative objects and then use only those objects as input for the algorithm. In this case, the work would be how to extend the results to all the objects on the original dataset. **There are several options in this format which rely basically on the combination of several models. You will learn then the strategy that you should use to combine learning models. Again here the project should contain all 4 parts with the focus on the impact of the combination of learning models on the results.**

## Expected Outcome

The expected outcome of the project is summarized as follows:
1. **A clear definition of the tasks** to be achieved and the characteristics of the chosen **dataset**.
2. **A principled approach in all the parts of the project.** All choices should be motivated. If the choice cannot be motivated at the beginning of the task, it should be followed by experiments that compare several alternatives.
3. **A project report (5-10 pages max)**
4. **An oral presentation of 15/20 minutes** where you describe all the work you have done.

# Deadlines

I. May 6th: Presentation of the project proposal (1 slide, ppt format)

II. June 1st: Submission on Teams (Project report in pdf, code)
*NB: in case of late submission respect to the deadline, a penalty of 1 point per day will be applied.*

III. June 3rd: Project presentations (20 mins + questions)